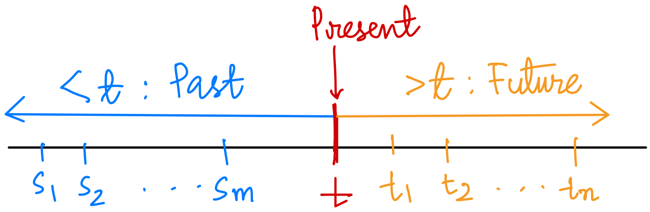
EE2080 : Lecture 8 : 12 Aug 2025



RECAP : 1) The Markov Property :

For an RP $\{X_t : t \in T\}$ on $(\Omega, \mathcal{F}, \mathbb{P})$,

$$\{X_{s_1}, X_{s_2}, \dots, X_{s_m}\} \perp\!\!\!\perp \{X_{t_1}, X_{t_2}, \dots X_{t_n}\} \,\Big|\, X_t \text{ , where}$$

$$s_1 < s_2 < \dots < s_m < t < t_1 < t_2 < \dots < t_n$$

Given the present, the future is independent of the past !

2) Discrete time Markov Chain

- $\{X_n, n \in \mathbb{N}\}$, with X_n taking values in a countable set S (State space) s.t $\{X_n, n\in \mathbb{N}\}$ has the Markov property
- Transition Probability matrix $\mathbb{P}(n) = |S| \times |S|$ matrix with the $(i,j)^{th}$ entry
  $P_{ij}(n) = \mathbb{P}\{X_{n+1} = j \mid X_n = i\}$, $n \in \mathbb{N}$, $i, j \in S$. (Row-Stochastic Matrix)
- Time Homogeneous DTMC $P(n) = P \;\; \forall\, n \in \mathbb{N}$
- n-step transition probabilities $\mathbb{P}\{X_{n+1} = j \mid X_1 = i\}$.

PLAN : 1) n-step transition probabilities – Chapman Kolmogorov Equation.

2) Finite dimensional distributions.

3) Strong Markov Property.

REFERENCE : A. Kumar, "Discrete Event Stochastic Processes".

n-step transition probabilities $\quad \mathbb{P}\{X_{n+1}=j \mid X_1=i\} = p_{i,j}^{(n)}$

i.e given that you are in state i, what is the prob. that you will be in state j after n time steps?

This is talked about only for time-homogeneous DTMCs.

Qn: For a time homogeneous DTMC $\{X_n : n \in \mathbb{N}\}$ on the state space S, is it required to specify the n-step transition probability matrix $P(n)$, with $p_{i,j}^{(n)} = \mathbb{P}\{X_{n+1}=j \mid X_1=i\}$ for all $n \in \mathbb{N}$?

Answer: NO!

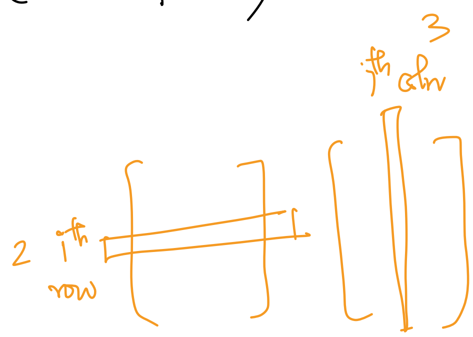
Consider $\{X_n, n \in \mathbb{N}\}$ DTMC with state space S.

$\{X_n, n\in\mathbb{N}\}$ is Time Homogeneous with TPM P.

$$\mathbb{P}\{X_{n+1}=j \mid X_n=i\} = p_{i,j} \quad \forall\, n \in \mathbb{N}$$

$$\mathbb{P}\{X_2=j \mid X_1=i\} = p_{i,j}$$

$$\begin{aligned}
\mathbb{P}\{X_3=j \mid X_1=i\} &= \sum_{k\in S} \mathbb{P}(X_3=j \mid X_2=k, X_1=i)\,\mathbb{P}(X_2=k \mid X_1=i) \\
&= \sum_{k\in S} \mathbb{P}(X_3=j \mid X_2=k)\,\mathbb{P}(X_2=k \mid X_1=i) \\
&= \sum_{k\in S} p_{k,j}\cdot p_{i,k} \\
&= (P^2)_{i,j}
\end{aligned}$$



We can get two-step transition probabilities from the one-step TPM!!

Chapman – Kolmogorov Equation

Theorem (Chapman – Kolmogorov): For a time-homogeneous DTMC $\{X_n : n \in \mathbb{N}\}$ with state space S, for any $n \in \mathbb{N}$, let $p_{i,j}^{(n)} = \mathbb{P}\{X_{n+1} = j \mid X_1 = i\}$, $i,j \in S$. Let $P^{(n)} = [p_{i,j}^{(n)}]_{i,j \in S}$. Then, $\boxed{P^{(n)} = P^n \ \forall n \in \mathbb{N}}$, where $P = P^{(1)}$ is the one-step TPM of $\{X_n, n \in \mathbb{N}\}$

Proof: For $i,j \in S$ and $n, l \geq 0$

$$
\begin{aligned}
p_{i,j}^{(n+l)} &= \mathbb{P}\{X_{n+l+1} = j \mid X_1 = i\} \\
&= \sum_{k \in S} \mathbb{P}\{X_{n+1} = k \mid X_1 = i\} \cdot P\{X_{n+l+1} = j \mid X_1 = i, X_{n+1} = k\} \\
&= \sum_{k \in S} p_{i,k}^{(n)} \cdot P\{X_{n+l+1} = j \mid X_{n+1} = k\} \quad \text{(Markov property)} \\
&= \sum_{k \in S} p_{i,k}^{(n)} \, p_{k,j}^{(l)} \\
&= \left[P^{(n)} P^{(l)} \right]_{i,j}
\end{aligned}
$$

$$\therefore P^{(n+l)} = P^{(n)} P^{(l)}.$$

Use this inductively to get the desired result.

Theorem (Chapman – Kolmogorov – II):

Let $\{X_n : n \in \mathbb{N}\}$ be a time-homogeneous DTMC on a state space S with TPM P. For each $n \in \mathbb{N}$, let $\pi(n)$ denote the unconditional PMF of X_n. Then

$$\boxed{\pi(n+1) = \pi(n) P}, \quad \forall n \in \mathbb{N}$$

Finite Dimensional distributions : Fix a prob. space $(\Omega, \mathcal{F}, \mathbb{P})$

FDDs of a time homogeneous DTMC $\{X_n, n \in \mathbb{N}\}$ on the state space S.

Consider, any $1 < n_1 < n_2 < \dots < n_m \in \mathbb{N}$ and $i_1, i_2, \dots, i_m \in S$,

$$\underbrace{\mathbb{P}(X_{n_1} = i_1, X_{n_2} = i_2, \dots X_{n_m} = i_m)}_{\text{Unconditional Joint Distribution}} = \sum_{i \in S} \mathbb{P}(X_1 = i, X_{n_1} = i_1, \dots, X_{n_m} = i_m)$$

$$= \sum_{i \in S} \mathbb{P}(X_1 = i)\, \mathbb{P}(X_{n_1} = i_1, \dots X_{n_m} = i_m \mid X_1 = i)$$

$$= \sum_{i \in S} \mathbb{P}(X_1 = i)\, \mathbb{P}(X_{n_1} = i_1 \mid X_1 = i)\, \mathbb{P}(X_{n_2} = i_2 \mid X_{n_1} = i_1) \cdots$$
$$\cdots \mathbb{P}(X_{n_m} = i_m \mid X_{n_{m-1}} = i_{m-1})$$

$$= \sum_{i \in S} \pi_i(1)\, P_{i, i_1}^{(n_1 - 1)} \cdot P_{i_1, i_2}^{(n_2 - n_1)} \cdots P_{i_{m-1}, i_m}^{(n_m - n_{m-1})}$$

$\therefore$ FDDs are completely specified using the TPM P and the initial pmf $\pi(1)$.

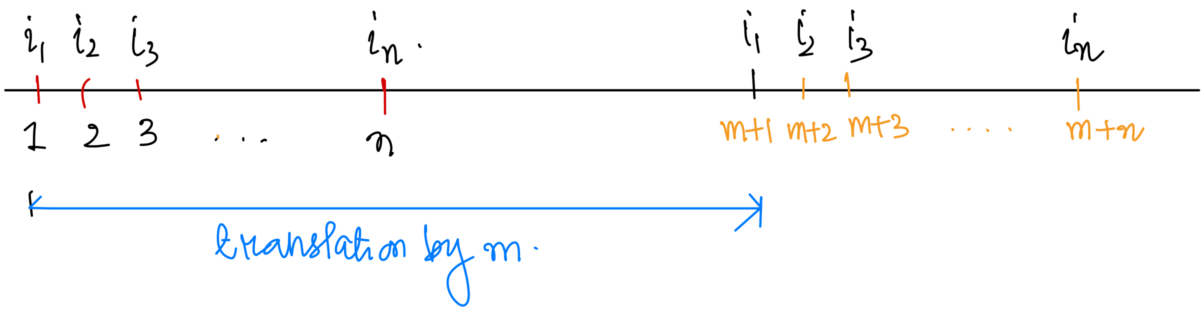
Remark: For a given TPM P, we obtain different DTMCs for different initial PMFs.

Stationarity of Conditional FDDs of TH DTMCs : Fix a prob. space $(\Omega, \mathcal{F}, \mathbb{P})$

Let $\{X_n : n \in \mathbb{N}\}$ be a time homogeneous DTMC on a state space S.

Conditioned on the initial state, any finite dimensional joint PMF of $\{X_n : n \in \mathbb{N}\}$ is stationary, i.e., for all $m, n \in \mathbb{N}$ and $i_1, i_2, \dots i_n \in S$,

$$\mathbb{P}\left(\bigcap_{j=2}^{n} \{X_j = i_j\} \,\middle|\, \{X_1 = i_1\}\right) = \mathbb{P}\left(\bigcap_{j=2}^{n} \{X_{m+j} = i_j\} \,\middle|\, \{X_{m+1} = i_1\}\right)$$



$$\mathbb{P}(X_2 = i_2, X_3 = i_3, \ldots, X_n = i_n \mid X_1 = i_1) = \mathbb{P}(X_2 = i_2 \mid X_1 = i_1) \cdot \mathbb{P}(X_3 = i_3 \mid X_2 = i_2) \ldots \mathbb{P}(X_n = i_n \mid X_{n-1} = i_{n-1})$$

$$= \prod_{j=2}^{n} \mathbb{P}(X_j = i_j \mid X_{j-1} = i_{j-1}) = \prod_{j=2}^{n} p_{i_{j-1}, i_j}$$

$$\mathbb{P}(X_{m+2} = i_2, \ldots, X_{m+n} = i_n \mid X_m = i_1) = \prod_{j=2}^{n} \mathbb{P}(X_{m+j} = i_j \mid X_{m+j-1} = i_{j-1}) = \prod_{j=2}^{n} p_{i_{j-1}, i_j}$$

In particular, for a TH-DTMC,

$$\boxed{\mathbb{P}\{X_{k+1} = j \mid X_1 = i\} = \mathbb{P}\{X_{k+m} = j \mid X_m = i\} \quad \forall\, i, j \in S,\ k, m \in \mathbb{N}}$$

Strong Markov Property: Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$

Suppose that $\{X_n : n \in \mathbb{N}\}$ is a time homogeneous DTMC on the state space S.

Markov Property: For any $n \in \mathbb{N}$

$$\mathbb{P}\{X_{n+1} = j \mid X_1 = i_1, \ldots X_{n-1} = i_{n-1}, X_n = i\} = \mathbb{P}\{X_{n+1} = j \mid X_n = i\}$$

i.e $X_{n+1} \perp\!\!\!\perp \{X_1, \ldots, X_{n-1}\} \mid X_n$

This is w.r.t a fixed n. Suppose T is a random time, i.e. T itself is a random variable taking values in the index set $\mathbb{N}$ of the DTMC, i.e $T : \Omega \longrightarrow \mathbb{N}$.

Does the Markov Property hold when n is replaced by T?

Example 1: Fix a prob. space $(\Omega, \mathcal{F}, \mathbb{P})$. Consider a TH DTMC $\{X_n : n \in \mathbb{N}\}$ on state space S. For $j \in S$, let T_j be the random time when the process first visits the state j, i.e $T_j(\omega) = k$ if $X_0(\omega) \neq j, X_1(\omega) \neq k, \ldots, X_{k-1}(\omega) \neq j, X_k(\omega) = j$

If for some $\omega \in \Omega$, $\nexists$ k s.t $T_j(\omega) = k$, then $T_j(\omega) = \infty$. Is T_j a r.v?

Is $\mathbb{P}(X_{T+k} = j \mid X_1 = i_1, \ldots, X_T = i) = p_{ij}^{(k)}$?